Neural Q-Function Approximation for an Episodic Stochastic Shortest-Path Problem

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***Abstract*—We replace the tabular Q-table from Assignment 2 with a feedforward neural network trained with a DQN-style update (experience replay, a target network, and a squared-error regression target) on the same eight-state stochastic shortest-path problem. The network never sees the transition model; it only trains on sampled transitions (s, a, c, s', d). Across five independent seeds (5000 episodes each), every trained network recovers the exact greedy policy found by tabular Q-learning and policy iteration in Assignment 2 ($a{\to}A_b$, $b{\to}A_e$, $c{\to}A_f$), with a 1000-episode stochastic evaluation mean cost within 0.4% of the known optimum of 8.44. We present the training curves, evaluation table, and a numerical hand-verification of one real forward pass, target, and TD error, and we argue that for a problem this small, tabular Q-learning remains the more practical choice; the neural approximator matches it but needs far more machinery to get there.**

I. Introduction

Assignment 2 solved an 8-state, undiscounted ($\gamma = 1$) episodic shortest-path MDP with a Q-table: one scalar per state-action pair, updated by a tabular Q-learning rule. A table does not scale to large or continuous state spaces, which is the usual motivation for replacing it with a parameterized function approximator. This assignment keeps the identical environment and cost-minimization objective and asks whether a small feedforward network, trained the way DQN trains on sampled transitions, can recover the same policy without ever seeing the transition probabilities or the Assignment-2 Q-table. We also ask the more practical question: on a problem this small, is the extra machinery (replay buffer, target network, gradient descent) actually worth it compared to the table?

II. Methodology

A. Environment

We reuse Assignment 2's environment exactly: states $\{a, b, c, d, e, f, g, h\}$, start a, terminal h with $V(h) = 0$, $\gamma = 1$. Branching states a, b, c each have two actions; the intended successor is reached w.p. 0.8 and the alternative branch w.p. 0.2 (Fig. 1). d, e, f, g have one action each. The realized cost is the edge cost of whichever branch actually occurs. The environment exposes only `step(s,a)` $\to (s', c)$; the learner never reads the transition-probability table or any tabular Q-values – those only reappear afterward, for comparison.

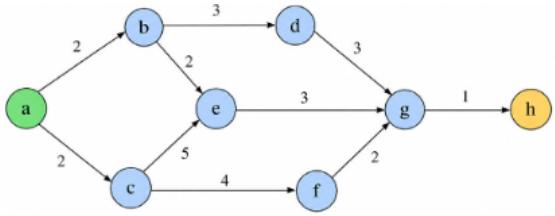


Fig. 1. Edge costs of the episodic shortest-path graph, identical to Assignment 2.

B. Neural Q-function

The state is one-hot encoded (8-dim). The network has two hidden layers of 32 ReLU units and one linear output per action (7 actions total across all states: $A_b, A_c, A_d, A_e, A_f, A_g, A_h$):

$$h_1 = \mathrm{ReLU}(W_1 s + b_1), \quad h_1 \in \mathbb{R}^{32} \tag{1}$$

$$h_2 = \mathrm{ReLU}(W_2 h_1 + b_2), \quad h_2 \in \mathbb{R}^{32} \tag{2}$$

$$Q_\theta(s, \cdot) = W_3 h_2 + b_3, \quad Q_\theta(s, \cdot) \in \mathbb{R}^7 \tag{3}$$

Actions infeasible at the current state are masked by adding $+10^6$ to their Q-value before any $\arg\min$ or $\min$ is taken – the cost-minimization analogue of $-\infty$ masking before a max. Trainable parameters: $W_1{\in}\mathbb{R}^{32\times 8}$ (256), b_1 (32), $W_2{\in}\mathbb{R}^{32\times 32}$ (1024), b_2 (32), $W_3{\in}\mathbb{R}^{7\times 32}$ (224), b_3 (7); **total** 1,575 trainable parameters, counted directly from the instantiated PyTorch module.

C. Learning target and loss

For a sampled transition (s, a, c, s', d), with $d{=}1$ iff $s'{=}h$, the target uses a separate target network Q_{θ^-}:

$$y = c + \gamma(1 - d) \min_{a' \in \mathcal{A}(s')} Q_{\theta^-}(s', a'), \tag{4}$$

which reduces to $y = c$ at a terminal transition (no bootstrap). The online network is trained by minibatch regression,

$$\mathcal{L}(\theta) = \frac{1}{B} \sum_{i=1}^{B} \big(y_i - Q_\theta(s_i, a_i)\big)^2, \tag{5}$$

with y_i detached from the autograd graph (semi-gradient: gradients flow through the prediction $Q_\theta(s_i, a_i)$, never through y_i), using Adam. **Experience replay** decorrelates consecutive gradient updates – every episode walks the same four-edge

TABLE I
HYPERPARAMETERS (IDENTICAL ACROSS ALL 5 SEEDS).

Parameter	Value
Hidden layers	2×32, ReLU
Optimizer	Adam, lr $= 10^{-3}$
Loss	MSE, Eq. (5)
Replay capacity / warm-up	10,000 / 200 transitions
Minibatch size	32
Target update	hard copy every 100 opt. steps
Discount γ	1.0
Episodes per seed	5000
ϵ-greedy schedule	$1.00 \rightarrow 0.05$ linearly, episodes 1–4000; 0.05 afte
Seeds	0, 1, 2, 3, 4

path, so training on transitions in the order they occur would feed the optimizer a narrow, correlated stream; sampling uniformly from a buffer breaks that correlation and reuses past transitions. **The target network** is held fixed between hard syncs so the regression target does not move on every gradient step the prediction also moves on, which is what keeps the bootstrapped regression from chasing itself.

D. Training protocol

Table I lists the hyperparameters, all at the assignment's suggested starting values, used unchanged for every seed. Each of the five seeds (0–4) gets a freshly constructed online network, target network (initialized equal to the online network), Adam optimizer, empty replay buffer, and its own `numpy.random.Generator`, with `random`, `numpy`, and `torch` global seeds reset to the seed value; no seed continues training from another.

III. RESULTS AND EVALUATION

A. Training curves (seed 4)

Fig. 2 shows seed 4's per-episode cost and its 100-episode rolling mean. The raw cost takes only the values $\{8, 9, 10, 11\}$ because this graph has exactly four edges on every a-to-h path and the per-episode cost is a sum of those discrete edge costs; the banding is a property of the environment, not noise in the learner. The rolling mean drops from about 9.2 early in training to settle near the 8.44 reference line (the Assignment-2 policy-iteration optimum) once ϵ reaches its 0.05 floor around episode 4000, with the usual residual spread from that exploration floor and the environment's own stochastic transitions.

Fig. 3 plots the training loss (log scale) with a 200-step rolling mean. The loss falls sharply over the first few hundred optimization steps as the network leaves its random initialization, then flattens to a noisy plateau around 0.1–0.2 for the rest of training. It does not fall toward zero, and we do not read that plateau as convergence by itself: with ϵ never going below 0.05 and the environment's own $0.8/0.2$ randomness, even a network that exactly matched the true optimal cost-to-go would still see nonzero squared error on every minibatch, because the realized cost c_i varies between samples of the same (s, a). The plateau is consistent with the network having reached a stable estimate of the mean bootstrap target, not with the loss itself vanishing.

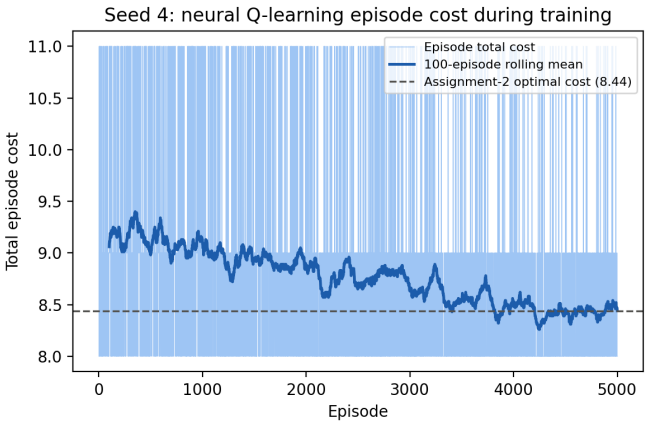


Fig. 2. Seed 4: per-episode total cost and its 100-episode rolling mean, against the Assignment-2 optimal cost (dashed).

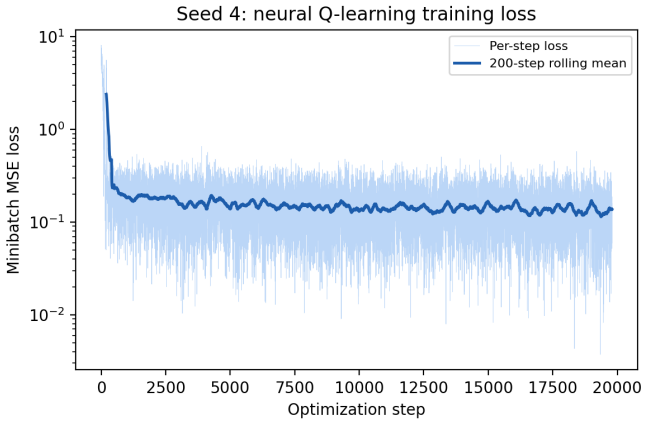


Fig. 3. Seed 4: training loss (log scale) with a 200-step rolling mean.

B. Learned policies and nominal paths

All five independently trained networks (seeds 0–4) recovered the same greedy policy at every one of the seven nonterminal states: $\pi(a){=}A_b$, $\pi(b){=}A_e$, $\pi(c){=}A_f$, $\pi(d){=}A_g$, $\pi(e){=}A_g$, $\pi(f){=}A_g$, $\pi(g){=}A_h$. This matches the optimal policy from both policy iteration and tabular Q-learning in Assignment 2. Under the intended transitions (the 0.8-probability branch at every decision), the nominal greedy path is $a \rightarrow b \rightarrow e \rightarrow g \rightarrow h$ for every seed – this is not the same as the expected stochastic path, since each branching step still has a 0.2 chance of landing on the alternate successor.

C. Stochastic evaluation (1000 episodes, $\epsilon = 0$)

Table II reports the frozen-network, exploration-off evaluation for each seed, using an evaluation RNG stream (seeded at $100{,}000{+}$seed) independent of the training RNG. Every seed reaches h in 100% of the 1000 episodes, with mean cost within 0.4% of the 8.44 optimum; the spread between the per-seed means (8.409–8.473) is well inside one standard deviation of a single seed's own evaluation noise (≈ 0.68), so it reads

TABLE II
FIVE-SEED STOCHASTIC EVALUATION, 1000 EPISODES/SEED, $\epsilon = 0$.

Seed	Mean	Std	Min	Max	% reach h
0	8.445	0.681	8.0	11.0	100.0
1	8.409	0.654	8.0	11.0	100.0
2	8.473	0.750	8.0	11.0	100.0
3	8.445	0.698	8.0	11.0	100.0
4	8.447	0.698	8.0	11.0	100.0
PI optimal $V^*(a)$			8.440		

as sampling variability, not a meaningfully different learned policy.

D. Hand verification

We verified one real sampled transition against the trained seed-4 network: $(s{=}b, a{=}A_e) \to c{=}2, s'{=}e, d{=}0$. The online network gives $Q_\theta(b, A_e){=}6.1291$; the target network (identical to the online network at the end of training) gives $\min_{a'} Q_{\theta^-}(e, a'){=}3.9649$ (the only feasible action at e is A_g, so the mask makes this exact). Substituting, $y = 2 + 1.0(1 - 0)(3.9649) = 5.9649$, TD error $\delta = 5.9649 - 6.1291 = -0.1642$, squared loss $\delta^2 = 0.0270$. Full forward-pass intermediate values (all 32 hidden units), a separate hand-computed illustrative 2-hidden-unit example (explicitly not the trained weights), and a real 8-transition minibatch loss check are in `verification.md`.

IV. Discussion

A. Neural vs. tabular Q-learning

Both methods see exactly the same thing during training: sampled (s, a, c, s', d) tuples, nothing more; neither ever touches `TRANSITIONS` directly. Where they differ is what they do with those samples. The tabular approach stores 10 Q-values, one per feasible state-action pair (two actions each at a, b, c, one each at d, e, f, g), and updates the corresponding entry after each transition. The network uses 1,575 trainable parameters to approximate those same 10 values, turning every transition into a gradient step through two ReLU layers – a lot of extra machinery, on an 8-state problem, to represent something a table already represents exactly.

Training stability tells a similar story. Assignment 2's tabular Q-learning converged to the exact optimal policy, zero gap, on all 5 baseline seeds, with nothing beyond a learning rate and an ϵ-schedule. The network needed a target network and a replay buffer just to keep the bootstrap target from chasing its own prediction, and even with both in place the loss in Fig. 3 never goes near zero; it plateaus around 0.1–0.2 and stays noisy for the rest of training. We do not read that as instability. The environment itself is stochastic and ϵ never drops below 0.05, so some irreducible variance in the targets is expected. But it does mean that a decreasing loss curve is not, by itself, evidence of convergence, which is the caution the assignment asks us not to skip.

On outcomes the two methods land in the same place. All five neural seeds recovered $a{\to}A_b$, $b{\to}A_e$, $c{\to}A_f$, matching tabular Q-learning and policy iteration exactly, and the evaluation means in Table II sit within about ± 0.03 of the 8.44 optimum, well inside a single seed's own evaluation noise – a wash between the two methods. The knob count is not: the table needs a learning rate and an exploration schedule, full stop; the network needs those plus an architecture, an optimizer and its learning rate, a replay capacity and warm-up length, a batch size, and a target-update period. Training all five neural seeds took about 30 seconds each on CPU (≈19,800 gradient steps per seed); the tabular baseline from Assignment 2 finishes in a fraction of that, with no PyTorch, no autograd graph, and no tensor overhead for what amounts to 10 scalar Q-values.

B. Is neural Q-learning beneficial here?

No, not for this problem. Eight states and seven actions fit a Q-table trivially, and tabular Q-learning already reaches the exact optimum with less code, no architecture choice, and less compute. The network reproduces the same answer – a useful sanity check that the implementation is correct – but it adds real complexity (replay buffer, target network, masking, extra hyperparameters) that buys nothing here. The case for a neural Q-function is the one Lecture 9 makes: states that cannot be enumerated or stored in a table at all, not a problem that already fits in one.

V. Conclusion

A small feedforward network trained with replay and a target network, on sampled transitions only, recovered the exact tabular-optimal policy and cost on this 8-state problem across all 5 independent seeds – good evidence the DQN-style update is implemented correctly. The main limitation is scale: an 8-state, 1575-parameter problem can't show what function approximation is actually for (generalizing across large or continuous state spaces), so nothing here says neural Q-learning scales well, only that on this particular small problem it reaches the same place tabular Q-learning does, at higher cost.

README / Reproducibility

Software: Python 3.13.9, PyTorch 2.15.0 (CPU build), NumPy 2.4.2, Matplotlib 3.10.8. **Hardware:** Windows 11, CPU-only, no GPU. **To run:** `cd code && python neural_q_learning.py` trains all 5 seeds (≈30 s/seed) and writes `results/` and `figures/`; `python hand_verification.py` regenerates the Section III-D numbers. Code and full results: https://github.com/Chinmay-Prashanth/eee598-neural-q-learning.

AI Assistance Disclosure

The team developed the solution, performed the experiments, and analyzed the results independently. Generative AI tools were used to assist with code optimization and to improve the vocabulary, grammar, and readability of the report. The final implementation and results were reviewed and verified by the team.